

Force Estimation for Admittance Control on Wuji Hand

23/04/2026

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Introduction

- Admittance control allows the robot to respond compliantly to external forces by modeling the system as a mass–spring–damper
- Goal: Achieve compliant admittance control on Wuji hand
- Problem:
 - Missing calibration/gain parameters (ηK_t)
 - Inaccurate estimated force vectors
 - Inconsistent current signal

Pipeline

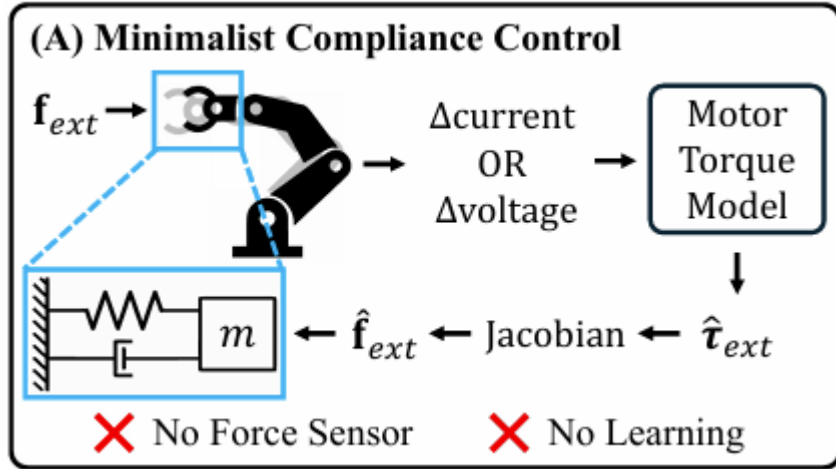


Figure from Minimalist Compliance Control

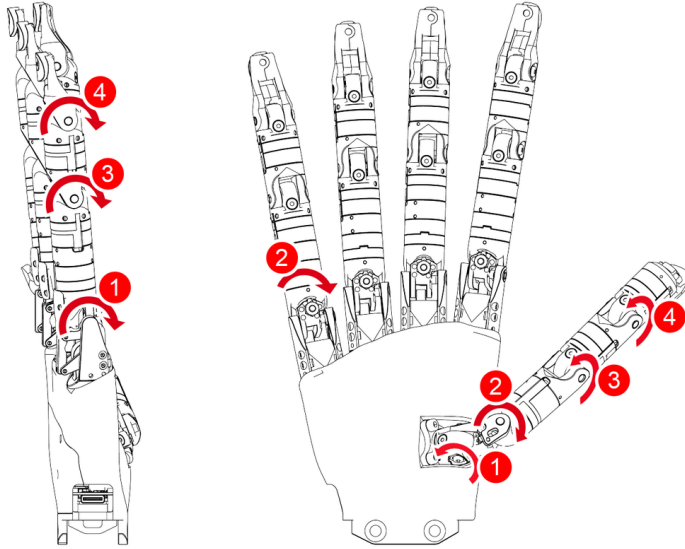
Motor torque from measured signals:

$$\tau = \eta K_t I_w$$

- τ - torque
- η - efficiency
- K_t - torque constant
- I_w - current

* Simplified algorithm that predicts the force in the contact direction

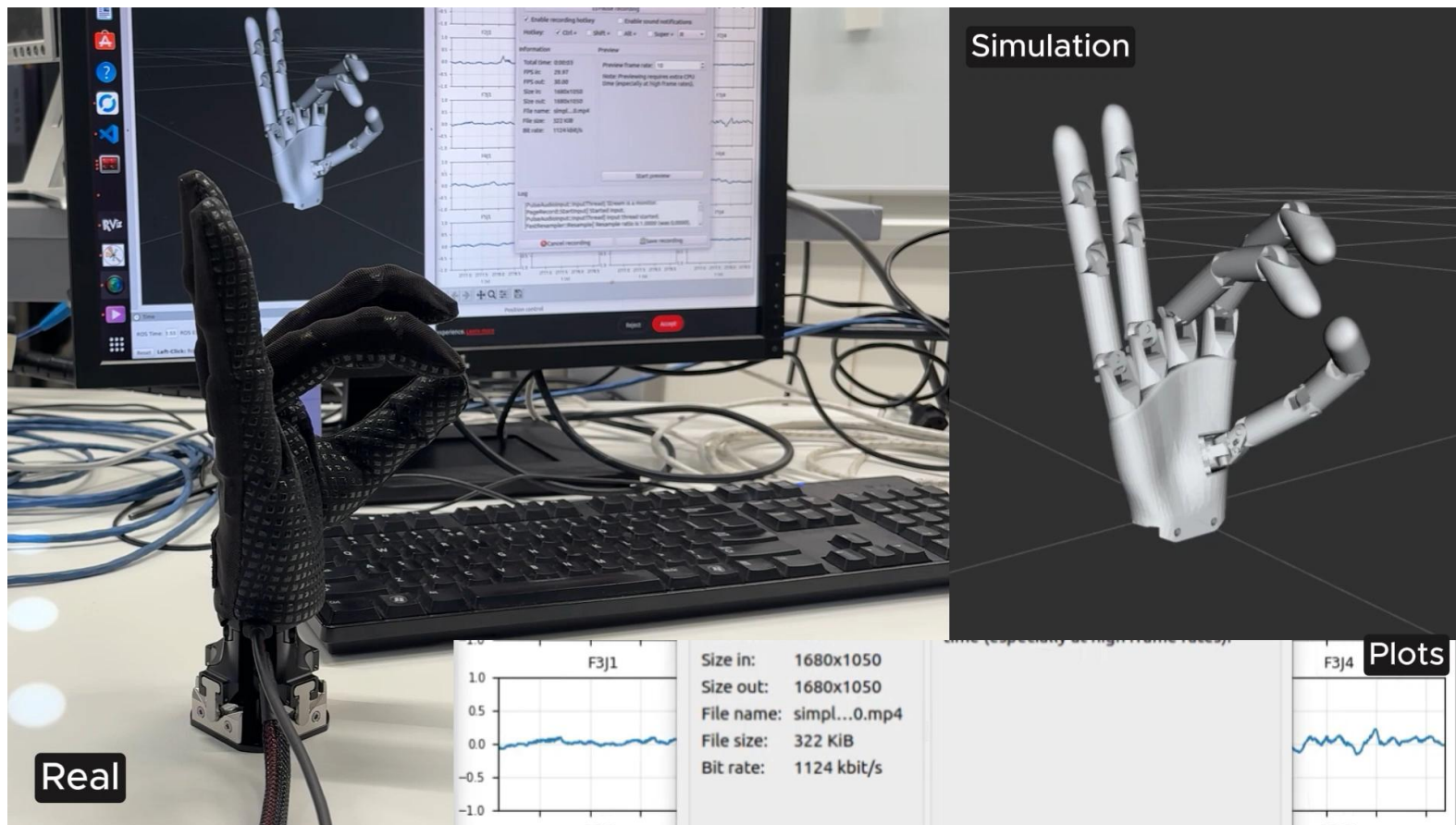
Wuji Hand



- **Joint 1:** Flexion/extension, motion range approximately 125°
- **Joint 2:** Lateral swing, motion range approximately 60°
- **Joint 3:** Flexion/extension, motion range approximately 130°
- **Joint 4:** Flexion/extension, motion range approximately 130°

Note: Joint 1 has a 3-link mechanism, not a motor that directly drives the joint

Data Recording



Observations

- Estimated force vector direction is inaccurate, especially along vertical axis
- Joint 2 (lateral) exhibits extreme currents magnitudes
- Possible coupling between joint 3 and 4

Next Steps

- Contact Wuji technical support for the expected gain (ηK_t) values for each joint, and how the behaviour of joint 1 is different under the linkage mechanism.
- Investigate possible joint coupling (J3 – J4)
- Improve torque estimation accuracy